STUDENT PERFORMANCE ANALYSIS

Domain Executive Briefing: Education Domain (student_performance.csv) | Date: August 2026

1. Executive Summary

Analysis of student academic performance, focusing on pass/fail rates, the impact of attendance on grades, and parental education influence.

2. Key Performance Indicators (KPI Overview)

Metric Description	Key Performance Value	Strategic Benchmark / Context
Overall Pass Rate	90.8%	Percentage of students passing
Avg Math Score	68.1	Average score across all students
Avg Attendance	81.7%	Overall attendance metric

3. Formal Statistical Hypothesis Testing

Parental Education Impact: --- FORMAL HYPOTHESIS TEST --- H■: Parent education is not associated with math scores. H■: Parent education is associated with math scores. Test used: One-Way ANOVA Test statistic (F-stat): 0.9397 p-value: 0.44009 95% Confidence Interval: N/A Effect Size (Eta Squared): 0.0038 Significance level: 0.05 Decision: Fail to Reject H■ Business interpretation: No significant evidence that parent education impacts math scores in this dataset. Why this matters: The result does not provide sufficient statistical evidence for a targeted intervention. ------------------------------

Attendance Correlation: --- FORMAL HYPOTHESIS TEST --- H■: No correlation between attendance and math scores. H■: Positive correlation between attendance and math scores. Test used: Pearson Correlation Test statistic (r): 0.4285 p-value: 0.00000 95% Confidence Interval: [0.3766, 0.4778] Significance level: 0.05 Decision: Reject H■ Business interpretation: High attendance correlates with high grades. Why this matters: Supports enforcing strict attendance policies. ------------------------------

4. Core Analytical & Statistical Insights

Insight 1: Finding: Higher attendance correlates positively with better scores (r=0.4285). Meaning: Class time is associated with success.

Insight 2: Finding: Students with different parental education levels show differences in average scores descriptively; however, the ANOVA does not provide sufficient evidence that these differences are statistically significant.

5. Graphical Analytics & Visual Evidence

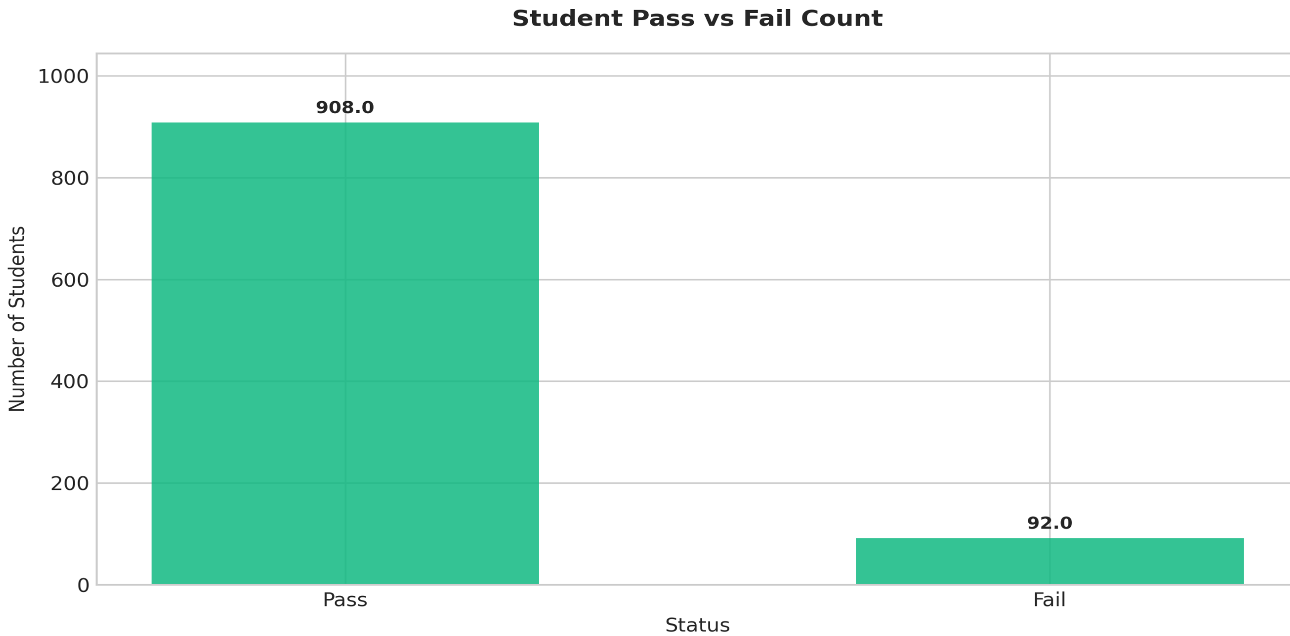

Figure 1: Visual analytical representation derived from Education Domain (student_performance.csv) dataset.

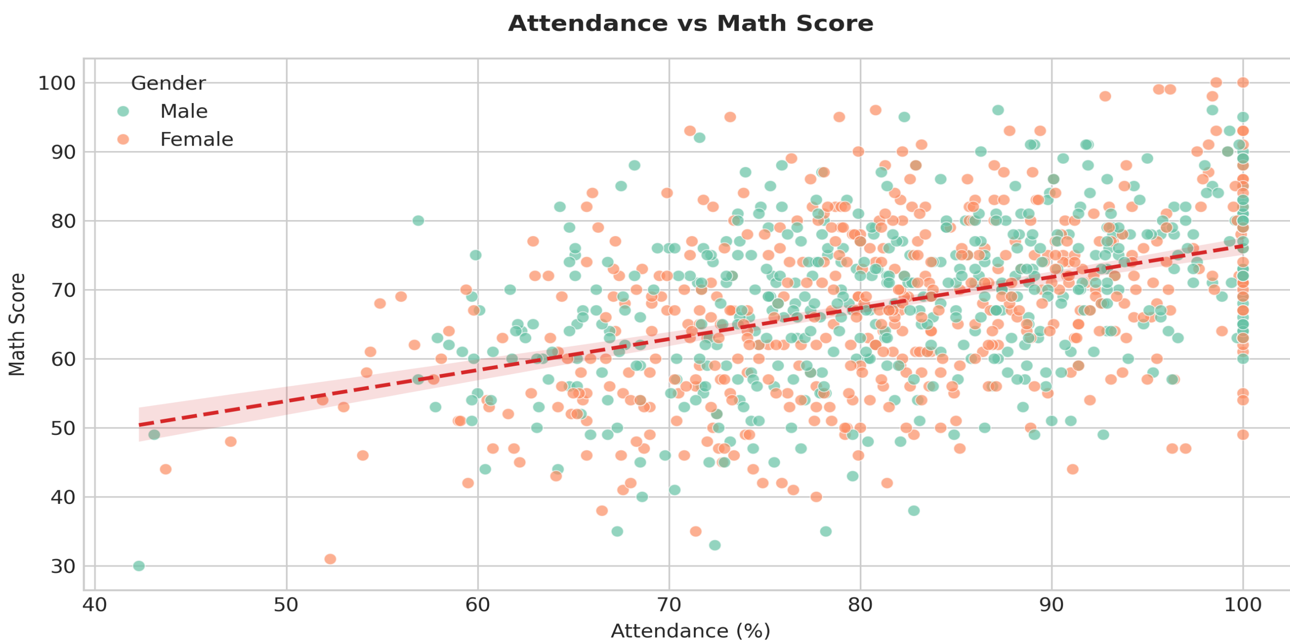

Figure 2: Visual analytical representation derived from Education Domain (student_performance.csv) dataset.

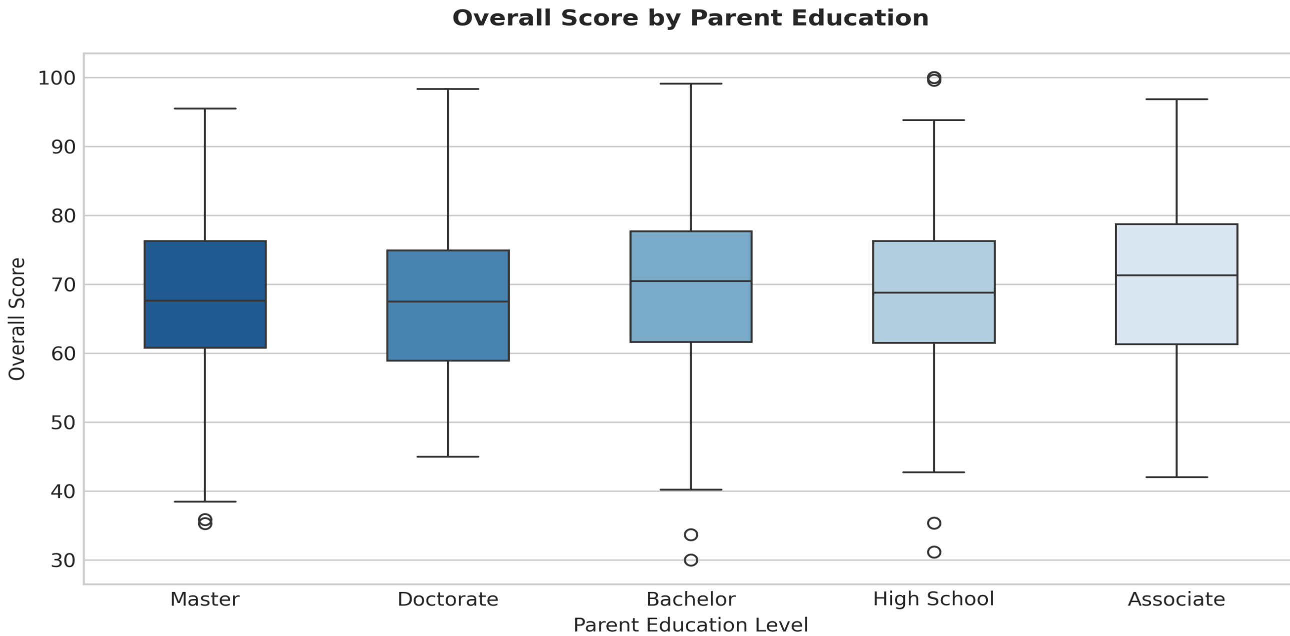

Figure 3: Visual analytical representation derived from Education Domain (student_performance.csv) dataset.

6. Strategic Recommendations & Action Plan

Action 1: Consider evaluating an early warning system for students dropping below 85% attendance.

Action 2: Evaluate adding after-school tutoring programs specifically aimed at first-generation students.